Loop Structure

Writing Loops

- **Loops:** repeated execution of one or more statements until a terminating condition occurs.

There are four types of loops:

- For … Next Statement.
- While … Statement.
- Do … Until Statement.
- Do … While Statement.

- **Writing For … Next Loops**
  - VB. NET For Next loop:
    - – Includes loop counter initialization and incrementing code as a part of the For statement.
    - – Uses pre-test logic – it evaluates the terminating expression at the beginning of the loop.
    - – You can use the For … Next statements to repeat a set of statements a specific number of times.

The syntax for the For … Next statement is as follows:

For Counter = <Start value> To <End Value> Step Value

Statement(s)

Next <Counter>

In the preceding syntax:

- Counter is any numeric variable.
- Start value is the initial value of Counter, and End value is the final or end value of Counter.
- Step value is the numeric value by which Counter needs to be incremented. The Step value can be either a positive or negative value and is optional. If no value is specified the default value is 1.
- Statement(s) are the set of statements executed for the specified number of times.
- The Next statement marks the end of a For … Next loop, then this statement is encountered, the Step value is added to Counter and the loop executes.

Example 1:

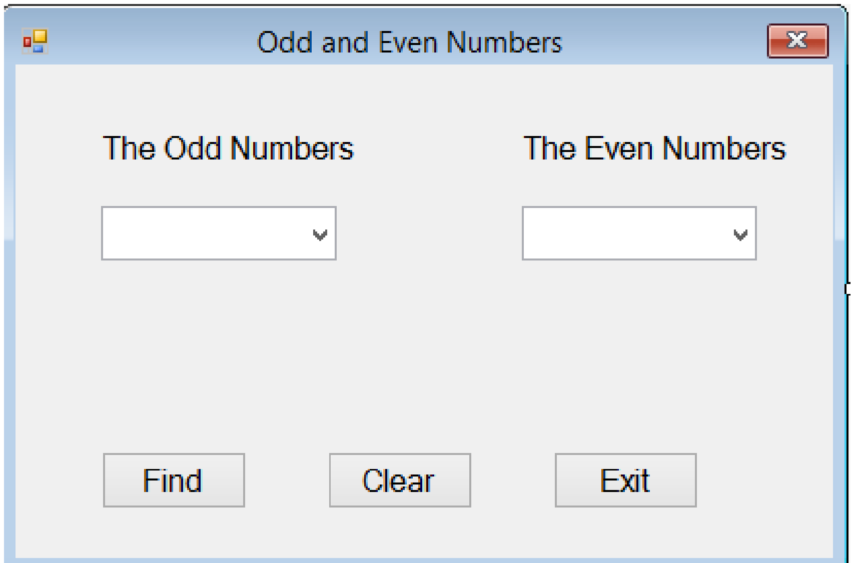
```
Dim Counter as Integer
    For Counter = 1 to 10 Step 2
        MessageBox.Show("Value is: " & Counter)
    Next Counter
```

Example 2:

```
Dim Num as Integer
    For Num = 0 to 10 Step 2
    MessageBox.Show(" The Even Number is " & Num)
    Next Num
```

Example 3:

Design the following form and write a program in VB.Net to read ten integer numbers then find the Odd and Even number between them using For … Next and ComboBox?



The Solution is:

To design the above form we implement the following set properties of the objects:

Object	Property	Setting
Button1	Text	Find
Button2	Text	Clear
Button3	Text	Exit
Label1	Text	The Odd Numbers
Labele2	Text	The Even Numbers
Form1	Text	Odd and Even Numbers
Form1	MaximizeBox	False
Forrm1	MinimizeBox	False
ComboBox1		
ComboBox2		

Writing the Code

Use the Code Editor

1. Double-click the Find Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim X As Integer
    Dim I As Integer
    For I = 1 To 10
      X = InputBox("Enter The 10 Numbers")
      If X Mod 2 = 0 Then
        ComboBox2.Items.Add(X)
      Else
        ComboBox1.Items.Add(X)
      End If
    Next
End Sub
```

2. Double-click the Clear Button2 on the form and write the following code inside it:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click

    ComboBox1.Items.Clear()

    ComboBox2.Items.Clear()

End Sub
```

3. Double-click the Exit Button3 on the form and write the following code inside it:

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        End
End Sub
```

- **Writing While … Statement**

The While … Statement is used to specify that a set of statements should repeat as long as the condition specified is True.

The syntax for the While … statement is as follows:

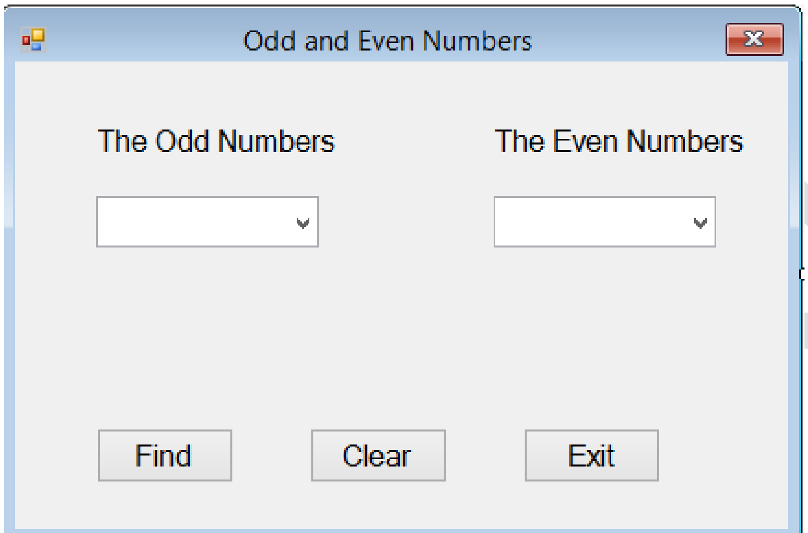
```
While Condition
        Statement(s)
Exit While
End While
```

Example 1:

```
Dim Counter as Integer
While Counter <=10
        MessageBox.Show("Value is :" & Counter)
Counter = Counter + 1
End While
```

Example2:

Design the following form and write a program in VB.Net to read ten integer numbers then find the Odd and Even number between them using While … Loop Statement and ComboBox?



The Solution is:

To design the above form we implement the following set properties of the objects:

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Button1	Text	Find
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Label1	Text	The Odd Numbers
Labele2	Text	The Even Numbers
Form1	Text	Odd and Even Numbers
Form1	MaximizeBox	False
Forrm1	MinimizeBox	False
ComboBox1		
ComboBox2		

Writing the Code

Use the Code Editor

1. Double-click the Find Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim X As Integer
    Dim I As Integer = 1
    While I <= 10
            X = InputBox("Enter The 10 Numbers")
            If X Mod 2 = 0 Then
            ComboBox2.Items.Add(X)
            Else
            ComboBox1.Items.Add(X)
            End If
      I = I + 1
    End While
   End Sub
```

2. Double-click the Clear Button2 on the form and write the following code inside it:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    ComboBox1.Items.Clear()
    ComboBox2.Items.Clear()
End Sub
```

3. Double-click the Exit Button3 on the form and write the following code inside it:

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    End
End Sub
```

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